N11: compressed sign-safe Kummer sidecar

25 October 2026

Status. An abstract differential-pair theorem supports one stored label when the iteration number is external. A bound for concrete theta xADD exceptions remains open. No cohort was run.

State and proof

For odd-prime-order generator G and signed public target T, store $(K(D), K(D+G), a, n)$ with $D = aG + 2^n T$. Branch $c \in \{0, 1\}$ sends D to $2D + cG$, a to $2a + c \pmod r$, and n to $n + 1$. This invariant follows by substitution. The new pair is respectively $(K(2D), K(2D+G))$ or $(K(2D+G), K(2D+2G))$.

If $(K(X), K(X+G)) = (K(Y), K(Y+G))$, then $Y = \pm X$. For $Y = -X$, equality of second coordinates implies $2X = 0$ (so $Y = X$) or $2G = 0$ (impossible). Hence the pair is injective, including at identity. After lifting its first coordinate, choose the sign with the second coordinate and verify $X = aG + (2^n \bmod r)T$ in the signed Jacobian.

Bounds and interpretation

There are exactly $\min(2^n, r)$ reachable labels after n unrestricted branch bits, requiring $\lceil \log_2 \min(2^n, r) \rceil$ bits in any exact local label state without the transcript. The concrete state is 72 bytes with an external step count, 80 bytes with a stored counter. The existing path uses three regular xADD calls, 72 field multiplications and 108 additions per step. The fixed-offset comparator has r explicit classes, or $32r$ bytes, but implements a different transition. The 336-row CSV retains every model and jump. Mathematical pair-sign ambiguities are zero; coordinate xADD exception bound and signed-lift cost are unknown. Cohort eligibility is false.